# Lab 3: Linear Regression

**NAME:**

In this lab, you will learn how to build and analyze linear regression models in R. The dataset we will use is a subset of the Ames Housing dataset (https://www.kaggle.com/c/house-prices-advanced-regression-techniques). Each row corresponds to a house sale in Ames, Iowa from 2006 to 2011. The variables we will use in this lab are

- **SalePrice** - final sale price of the house
- **LotArea** - lot size in square feet
- **Neighborhood** - categorical variable for neighborhood
- **OverallQual** - overall material and finishing quality rating
- **OverallCond** - overall condition rating
- **YearBuilt** - year the house was built
- **GrLivArea** - above ground living area in square feet
- **BedroomAbvGr** - total number of bedrooms above ground
- **BathAbvGr** - total number of baths above ground

Load some familiar libraries and the data set ames_house_sub.csv, which can be found in the Data folder on Canvas.

```
library(dplyr)
```

```
Attaching package: 'dplyr'

The following objects are masked from 'package:stats':

    filter, lag
```

```
The following objects are masked from 'package:base':

    intersect, setdiff, setequal, union
```

```r
library(ggplot2)
library(Hmisc)
```

```
Attaching package: 'Hmisc'

The following objects are masked from 'package:dplyr':

    src, summarize

The following objects are masked from 'package:base':

    format.pval, units
```

```r
library(GGally)
house.sales = read.csv("ames_house_sub.csv")
house.sales$Neighborhood = as.factor(house.sales$Neighborhood)
house.sales = subset(house.sales, is.finite(SalePrice) & !is.na(Neighborhood))
```

We first note that the dataset is complete so we do not have to deal with any missing data.

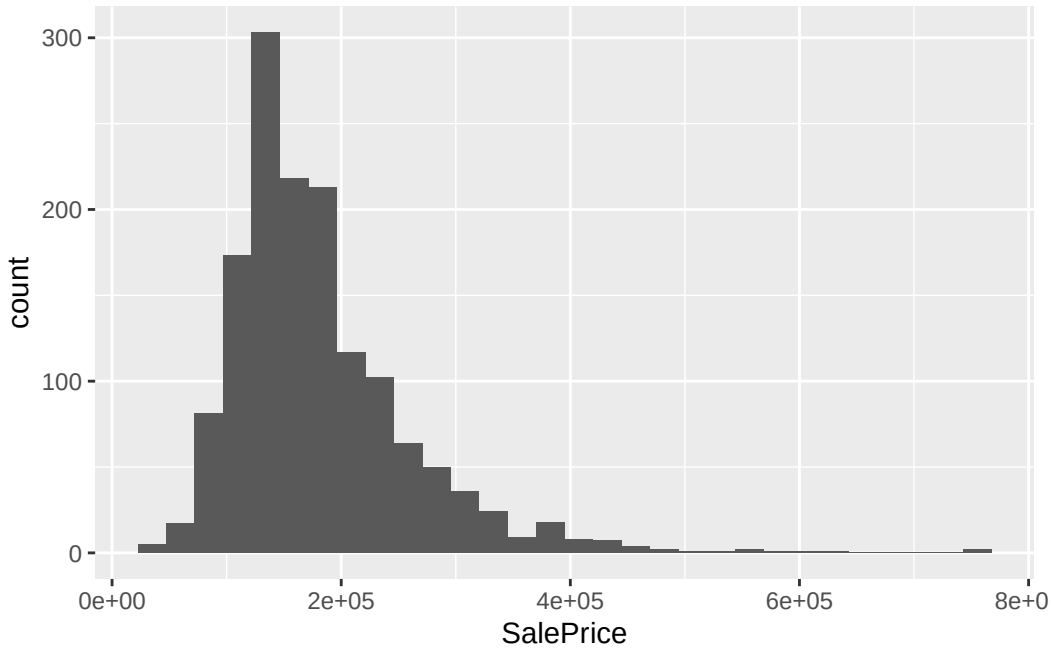
```r
sum(!complete.cases(house.sales))
```

```
[1] 0
```

The response variable we will be interested in predicting is sale price. Plot a histogram for this variable using ggplot. You should notice that this variable is fairly right-skewed. Plotting a histogram for log(SalePrice) gives a more balanced distribution. This might be useful later on since highly skewed distributions can lead to large outliers in a model.

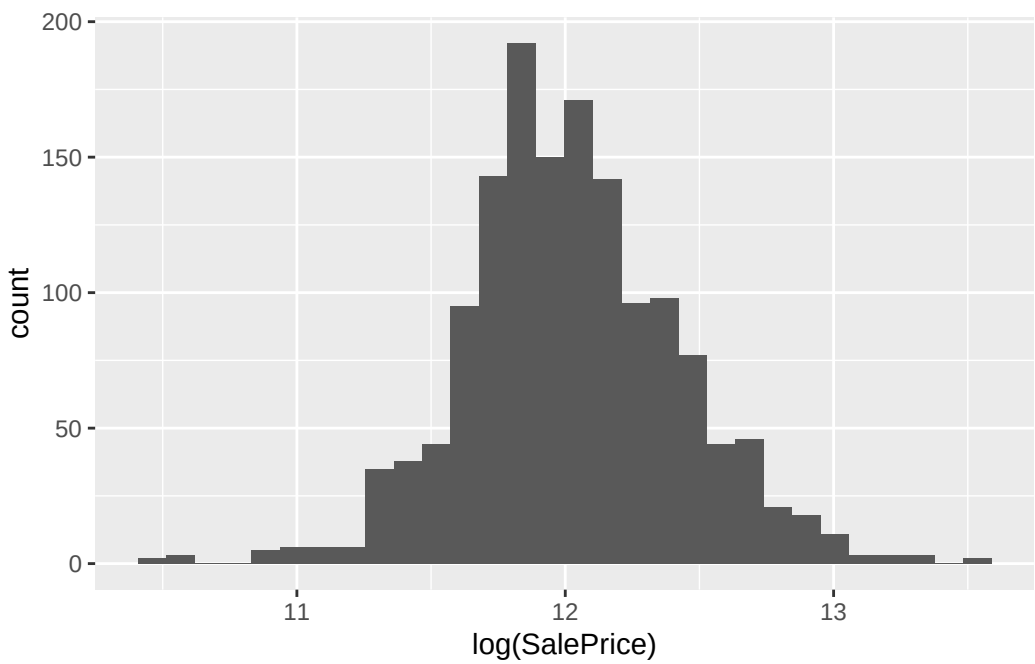
```r
ggplot(data=house.sales) + geom_histogram(aes(x=SalePrice))
```

```
`stat_bin()` using `bins = 30`. Pick better value `binwidth`.
```




```
ggplot(data=house.sales) + geom_histogram(aes(x=log(SalePrice)))
```

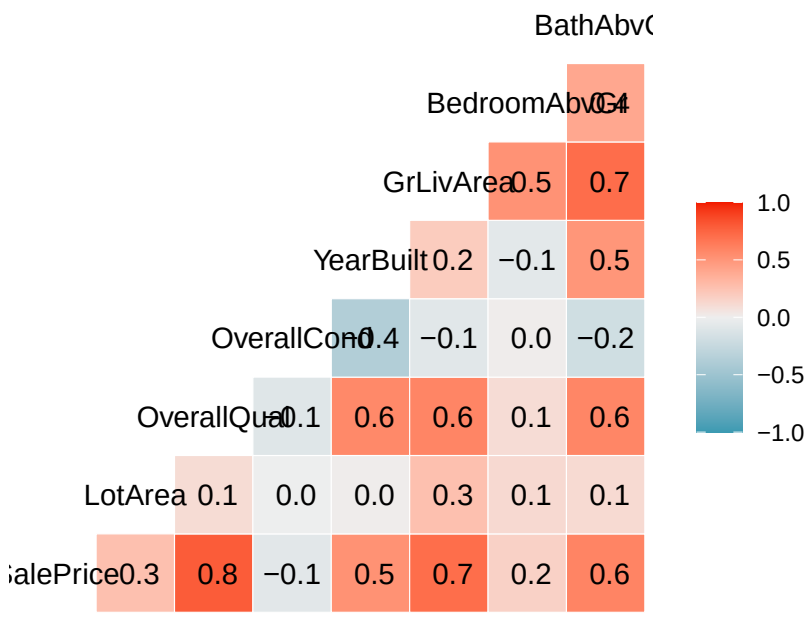
```
`stat_bin()` using `bins = 30`. Pick better value `binwidth`.
```



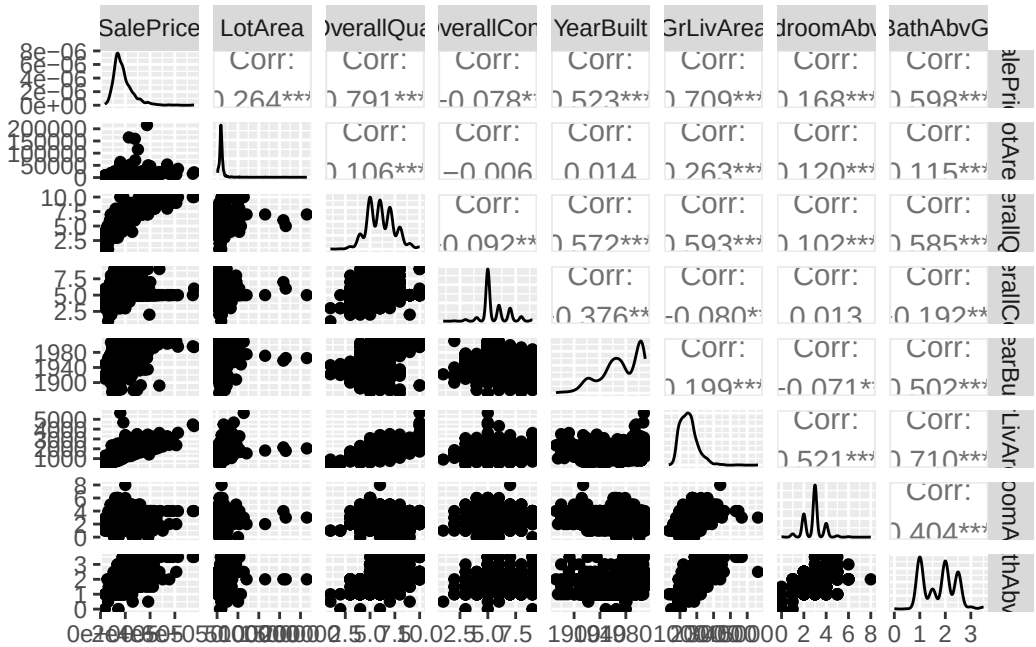
On your own, explore the other variable distributions using **describe or summary** and by plotting the distributions using histograms or bar plots. Some things that you should notice include that there are some very large outliers for LotArea, GrLivArea appears to be right-skewed, and OverallCond is set to 5 for the majority of houses (this might be the default value).

To explore the relationships between these variables, we will use the package GGally. Install and load this package and then run the following commands.

```
house.sales.sub = house.sales %>% select(-Neighborhood)
ggcorr(house.sales.sub,label=TRUE)
```



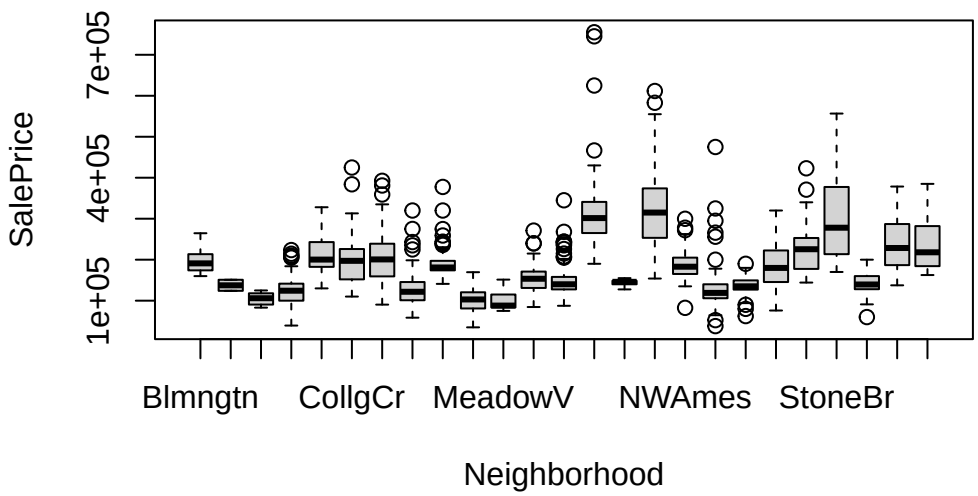

```
ggpairs(house.sales.sub)
```




ggcorr gives the correlation matrix between all continuous variables and ggpairs gives the correlations along with the corresponding plots. What do you notice when you look at these outputs? Which variables are highly correlated with SalePrice? Do the relationships look linear?

Since Neighborhood is a categorical variable, we can use boxplots to understand the relationship with SalePrice.

```
plot(SalePrice ~ Neighborhood, data = house.sales)
```




Here, we notice that there are a few neighborhoods that have much higher SalePrice values. # Simple Linear Regression We will start with a simple linear regression model using OverallQual, the variable that is most highly correlated with SalePrice. To build a linear model we use the command lm.

The first part specifies the model formula where the left-hand side is the response variable and the right-hand side has the predictors. The second part specifies the data. The output lm.simple gives us a lot of information. Find the different output values using **names**.

```
lm.simple <- lm(SalePrice ~ OverallQual, data = house.sales)
names(lm.simple)
```

```
 [1] "coefficients"  "residuals"     "effects"       "rank"
 [5] "fitted.values" "assign"        "qr"            "df.residual"
 [9] "xlevels"       "call"          "terms"         "model"
```

We can see that this includes the coefficient estimates, the fitted values, and the residuals. For a succinct summary of the model we can use the **summary** function.

```
summary(lm.simple)
```

```
Call:
lm(formula = SalePrice ~ OverallQual, data = house.sales)

Residuals:
    Min      1Q  Median      3Q     Max 
-198152  -29409   -1845   21463  396848 

Coefficients:
            Estimate Std. Error t value Pr(>|t|)    
(Intercept) -96206.1     5756.4  -16.71   <2e-16 ***
OverallQual  45435.8      920.4   49.36   <2e-16 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 48620 on 1458 degrees of freedom
Multiple R-squared:  0.6257,	Adjusted R-squared:  0.6254 
F-statistic:  2437 on 1 and 1458 DF,  p-value: < 2.2e-16
```

When looking at the summary, try to answer the following questions:

What is the $R^2$ and Adjusted $R^2$ value? How does this relate to the correlation between OverallQual and SalePrice?

How do you interpret the $R^2$ value? Does this seem like a good fit to you?

How would you interpret the two coefficient values (the intercept and the coefficient for OverallQual)?

What do the Std. Error and p-values for the coefficients represent?

Another useful function is the **predict** function. This uses the linear model to predict house prices.

```r
predict(lm.simple,data=house.sales)
```

```
         1          2          3          4          5          6          7 
221844.539 176408.736 221844.539 221844.539 267280.341 130972.933 267280.341 
         8          9         10         11         12         13         14 
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        15         16         17         18         19         20         21 
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        22         23         24         25         26         27         28 
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        29         30         31         32         33         34         35 
```

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| 130972.933 | 85537.131 | 85537.131 | 130972.933 | 267280.341 | 130972.933 | 312716.144 |
| 36 | 37 | 38 | 39 | 40 | 41 | 42 |
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| 43 | 44 | 45 | 46 | 47 | 48 | 49 |
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| 50 | 51 | 52 | 53 | 54 | 55 | 56 |
| 130972.933 | 176408.736 | 176408.736 | 130972.933 | 312716.144 | 130972.933 | 176408.736 |
| 57 | 58 | 59 | 60 | 61 | 62 | 63 |
| 267280.341 | 221844.539 | 358151.946 | 130972.933 | 176408.736 | 130972.933 | 267280.341 |
| 64 | 65 | 66 | 67 | 68 | 69 | 70 |
| 221844.539 | 221844.539 | 267280.341 | 221844.539 | 221844.539 | 85537.131 | 221844.539 |
| 71 | 72 | 73 | 74 | 75 | 76 | 77 |
| 221844.539 | 85537.131 | 221844.539 | 130972.933 | 40101.328 | 85537.131 | 85537.131 |
| 78 | 79 | 80 | 81 | 82 | 83 | 84 |
| 130972.933 | 85537.131 | 130972.933 | 176408.736 | 176408.736 | 267280.341 | 130972.933 |
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| 221844.539 | 267280.341 | 176408.736 | 176408.736 | 40101.328 | 85537.131 | 85537.131 |
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| 130972.933 | 85537.131 | 176408.736 | 176408.736 | 130972.933 | 221844.539 | 221844.539 |
| 106 | 107 | 108 | 109 | 110 | 111 | 112 |
| 267280.341 | 85537.131 | 130972.933 | 130972.933 | 176408.736 | 176408.736 | 221844.539 |
| 113 | 114 | 115 | 116 | 117 | 118 | 119 |
| 221844.539 | 176408.736 | 176408.736 | 176408.736 | 130972.933 | 130972.933 | 221844.539 |
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| 127 | 128 | 129 | 130 | 131 | 132 | 133 |
| 176408.736 | 130972.933 | 176408.736 | 130972.933 | 221844.539 | 176408.736 | 130972.933 |
| 134 | 135 | 136 | 137 | 138 | 139 | 140 |
| 267280.341 | 130972.933 | 221844.539 | 130972.933 | 221844.539 | 267280.341 | 176408.736 |
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| 85537.131 | 221844.539 | 130972.933 | 221844.539 | 130972.933 | 176408.736 | 130972.933 |
| 148 | 149 | 150 | 151 | 152 | 153 | 154 |
| 221844.539 | 221844.539 | 130972.933 | 130972.933 | 267280.341 | 176408.736 | 176408.736 |
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| 176408.736 | 176408.736 | 130972.933 | 267280.341 | 221844.539 | 221844.539 | 176408.736 |
| 162 | 163 | 164 | 165 | 166 | 167 | 168 |
| 312716.144 | 221844.539 | 85537.131 | 176408.736 | 130972.933 | 130972.933 | 267280.341 |
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| 221844.539 | 267280.341 | 130972.933 | 176408.736 | 221844.539 | 176408.736 | 176408.736 |
| 176 | 177 | 178 | 179 | 180 | 181 | 182 |
| 176408.736 | 176408.736 | 130972.933 | 312716.144 | 130972.933 | 176408.736 | 221844.539 |

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| 85537.131 | 221844.539 | 221844.539 | 221844.539 | 176408.736 | 176408.736 | 85537.131 |
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| 267280.341 | 176408.736 | 267280.341 | 130972.933 | 130972.933 | 176408.736 | 221844.539 |
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| 85537.131 | 130972.933 | 176408.736 | 267280.341 | 130972.933 | 130972.933 | 221844.539 |
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| 221844.539 | 176408.736 | 130972.933 | 267280.341 | 176408.736 | 221844.539 | 221844.539 |
| 428 | 429 | 430 | 431 | 432 | 433 | 434 |
| 85537.131 | 221844.539 | 176408.736 | 176408.736 | 176408.736 | 130972.933 | 176408.736 |
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| 85537.131 | 221844.539 | 176408.736 | 176408.736 | 130972.933 | 176408.736 | 358151.946 |
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| 85537.131 | 130972.933 | 221844.539 | 221844.539 | 176408.736 | 176408.736 | 221844.539 |
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| 176408.736 | 267280.341 | 221844.539 | 176408.736 | 267280.341 | 267280.341 | 130972.933 |
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| 176408.736 | 130972.933 | 130972.933 | 130972.933 | 130972.933 | 130972.933 | 85537.131 |
| 491 | 492 | 493 | 494 | 495 | 496 | 497 |
| 130972.933 | 176408.736 | 176408.736 | 130972.933 | 130972.933 | 85537.131 | 267280.341 |
| 498 | 499 | 500 | 501 | 502 | 503 | 504 |
| 221844.539 | 130972.933 | 130972.933 | 176408.736 | 221844.539 | 130972.933 | 221844.539 |
| 505 | 506 | 507 | 508 | 509 | 510 | 511 |
| 176408.736 | 130972.933 | 267280.341 | 176408.736 | 221844.539 | 130972.933 | 130972.933 |
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| 221844.539 | 130972.933 | 176408.736 | 130972.933 | 358151.946 | 176408.736 | 221844.539 |
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| 221844.539 | 130972.933 | 312716.144 | 85537.131 | 176408.736 | 176408.736 | 176408.736 |
| 533 | 534 | 535 | 536 | 537 | 538 | 539 |
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| 267280.341 | 312716.144 | 267280.341 | 221844.539 | 221844.539 | 221844.539 | 221844.539 |
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| 85537.131 | 221844.539 | 130972.933 | 130972.933 | 85537.131 | 221844.539 | 221844.539 |
| 561 | 562 | 563 | 564 | 565 | 566 | 567 |
| 130972.933 | 130972.933 | 130972.933 | 176408.736 | 221844.539 | 176408.736 | 312716.144 |
| 568 | 569 | 570 | 571 | 572 | 573 | 574 |
| 221844.539 | 267280.341 | 130972.933 | 130972.933 | 176408.736 | 221844.539 | 221844.539 |
| 575 | 576 | 577 | 578 | 579 | 580 | 581 |
| 130972.933 | 130972.933 | 221844.539 | 130972.933 | 221844.539 | 130972.933 | 176408.736 |
| 582 | 583 | 584 | 585 | 586 | 587 | 588 |
| 267280.341 | 176408.736 | 358151.946 | 85537.131 | 267280.341 | 176408.736 | 130972.933 |
| 589 | 590 | 591 | 592 | 593 | 594 | 595 |
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| 596 | 597 | 598 | 599 | 600 | 601 | 602 |
| 267280.341 | 176408.736 | 221844.539 | 130972.933 | 176408.736 | 267280.341 | 176408.736 |
| 603 | 604 | 605 | 606 | 607 | 608 | 609 |
| 267280.341 | 221844.539 | 221844.539 | 221844.539 | 130972.933 | 130972.933 | 267280.341 |
| 610 | 611 | 612 | 613 | 614 | 615 | 616 |
| 85537.131 | 312716.144 | 176408.736 | 267280.341 | 130972.933 | 85537.131 | 176408.736 |
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| 176408.736 | 176408.736 | 312716.144 | 267280.341 | 40101.328 | 176408.736 | 130972.933 |
| 624 | 625 | 626 | 627 | 628 | 629 | 630 |
| 176408.736 | 176408.736 | 176408.736 | 130972.933 | 176408.736 | 130972.933 | 176408.736 |
| 631 | 632 | 633 | 634 | 635 | 636 | 637 |

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| 130972.933 | 130972.933 | 176408.736 | 130972.933 | 130972.933 | 130972.933 | 221844.539 |
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| 221844.539 | 130972.933 | 267280.341 | 267280.341 | 221844.539 | 221844.539 | 221844.539 |
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| 85537.131 | 85537.131 | 221844.539 | 176408.736 | 267280.341 | 176408.736 | 176408.736 |
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| 267280.341 | 176408.736 | 221844.539 | 267280.341 | 176408.736 | 130972.933 | 221844.539 |
| 764 | 765 | 766 | 767 | 768 | 769 | 770 |
| 267280.341 | 267280.341 | 312716.144 | 221844.539 | 176408.736 | 221844.539 | 267280.341 |
| 771 | 772 | 773 | 774 | 775 | 776 | 777 |
| 130972.933 | 85537.131 | 176408.736 | 130972.933 | 267280.341 | 176408.736 | 221844.539 |
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| 130972.933 | 130972.933 | 176408.736 | 221844.539 | 176408.736 | 221844.539 | 130972.933 |

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| 130972.933 | 176408.736 | 130972.933 | 221844.539 | 130972.933 | 267280.341 | 176408.736 |
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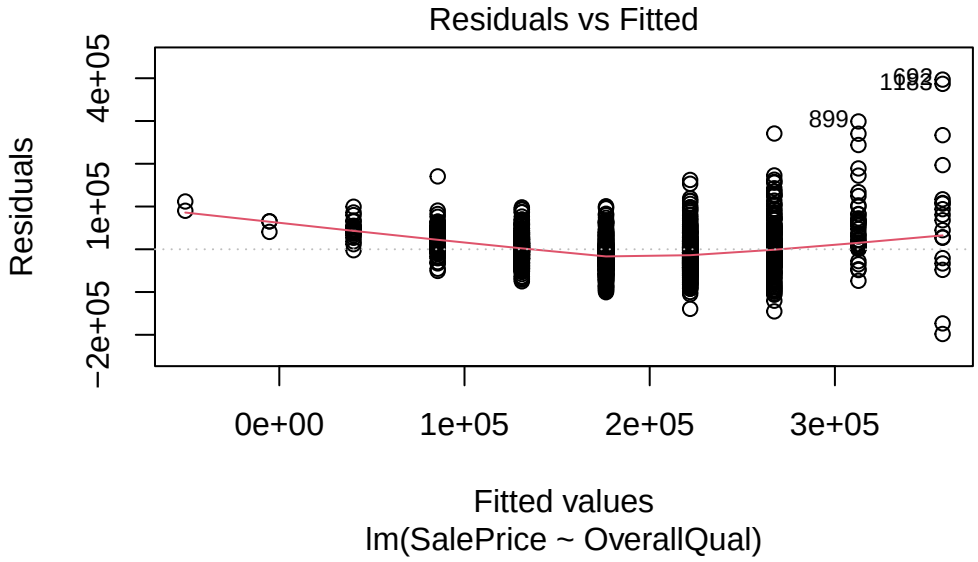
176408.736  85537.131 267280.341 176408.736 221844.539 176408.736 176408.736
      1443       1444       1445       1446       1447       1448       1449
358151.946 176408.736 221844.539 176408.736 130972.933 267280.341  85537.131
      1450       1451       1452       1453       1454       1455       1456
130972.933 130972.933 267280.341 130972.933 130972.933 221844.539 176408.736
      1457       1458       1459       1460
176408.736 221844.539 130972.933 130972.933
```

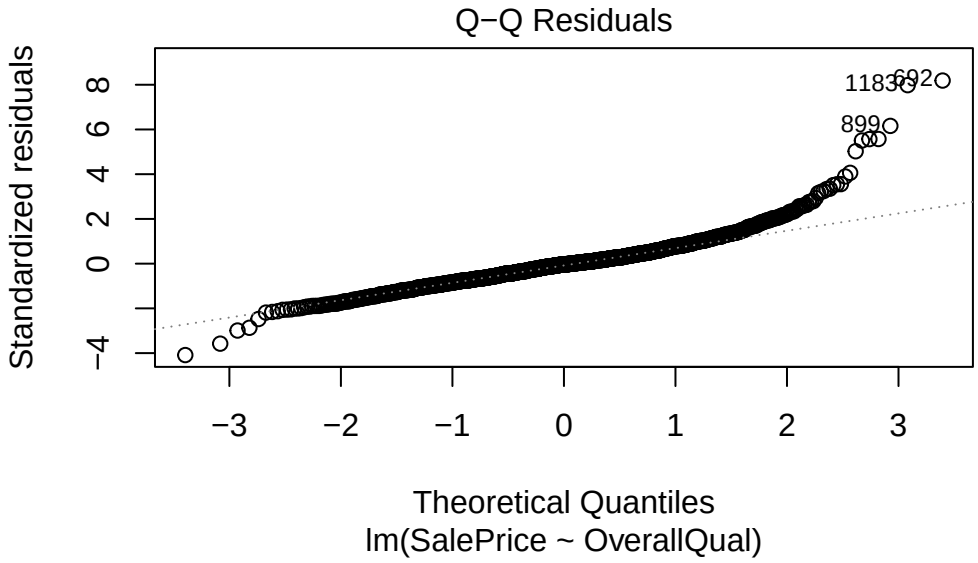
Above, we have found the predicted values on the training set, but by changing the input to data we could also predict on a test dataset.

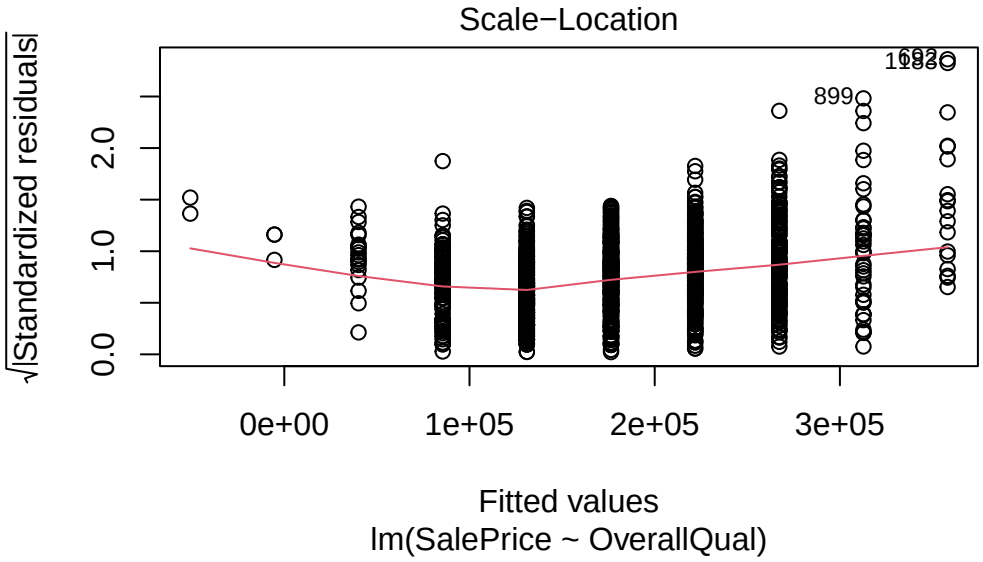
## Diagnostic Plots

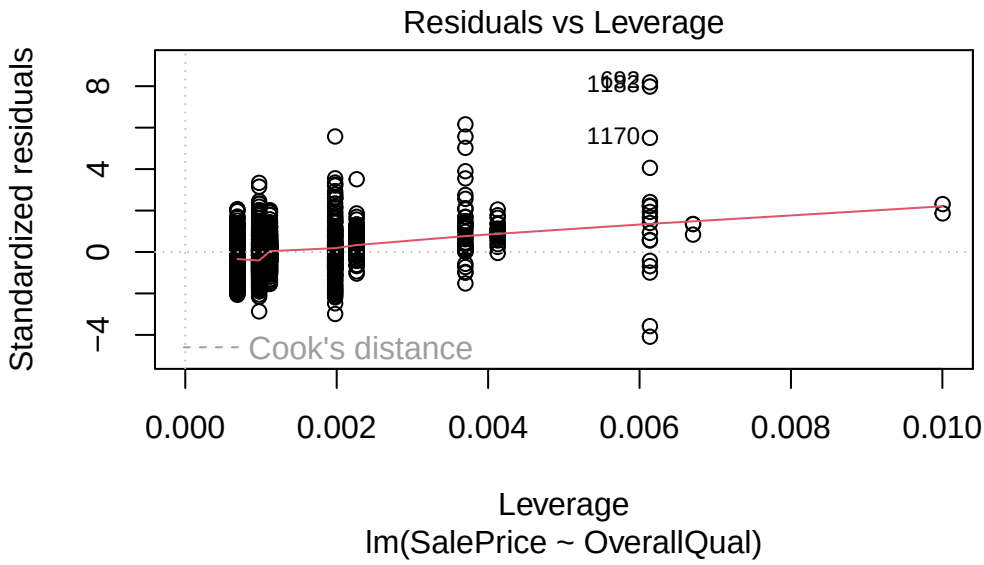
Having fit the model, we now want to run some diagnostic tests. There are two easy ways to generate some key diagnostic plots - using the plot function or using the ggfortify package and autoplot function. Try out both and observe the outputted plots (you can ignore the Scale-Leverage plot which we did not talk about in class).

```r
plot(lm.simple)
```


Residuals vs Fitted
Residuals
Fitted values
lm(SalePrice ~ OverallQual)
899
692
1183
−2e+05
1e+05
4e+05
0e+00
1e+05
2e+05
3e+05


Q−Q Residuals
Standardized residuals
Theoretical Quantiles
lm(SalePrice ~ OverallQual)
1183
692
899
−4
0
2
4
6
8
−3
−2
−1
0
1
2
3



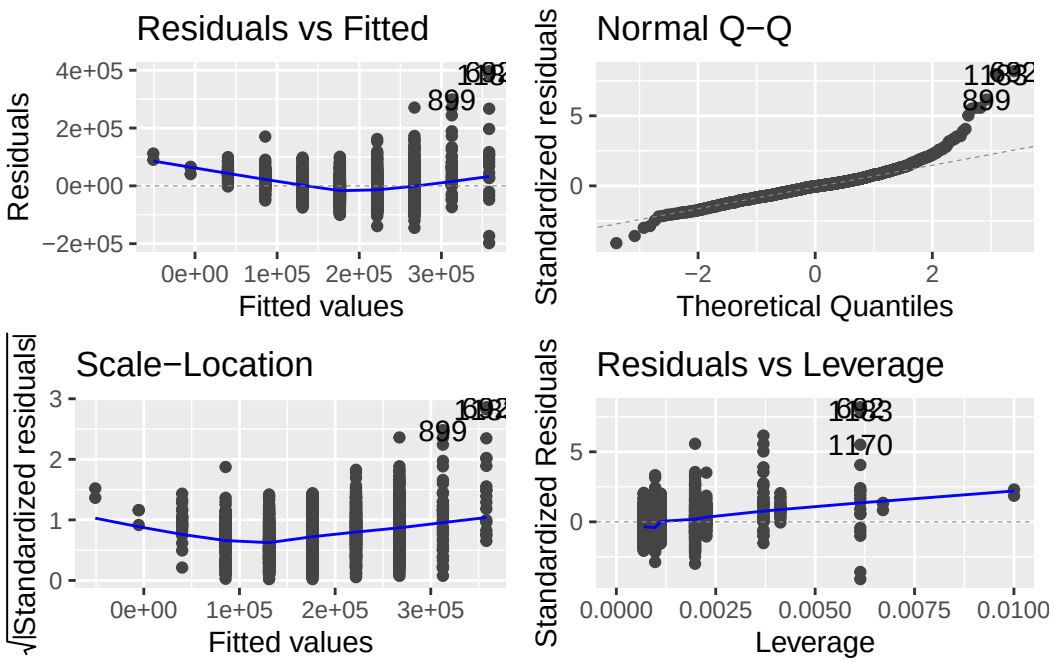

```
library(ggfortify)
autoplot(lm.simple)
```

```
Warning: `fortify(<lm>)` was deprecated in ggplot2 4.0.0.
```

```
i Please use `broom::augment(<lm>)` instead.
i The deprecated feature was likely used in the ggfortify package.
  Please report the issue at <https://github.com/sinhrks/ggfortify/issues>.

Warning: `aes_string()` was deprecated in ggplot2 3.0.0.
i Please use tidy evaluation idioms with `aes()`.
i See also `vignette("ggplot2-in-packages")` for more information.
i The deprecated feature was likely used in the ggfortify package.
  Please report the issue at <https://github.com/sinhrks/ggfortify/issues>.

Warning: Using `size` aesthetic for lines was deprecated in ggplot2 3.4.0.
i Please use `linewidth` instead.
i The deprecated feature was likely used in the ggfortify package.
  Please report the issue at <https://github.com/sinhrks/ggfortify/issues>.
```



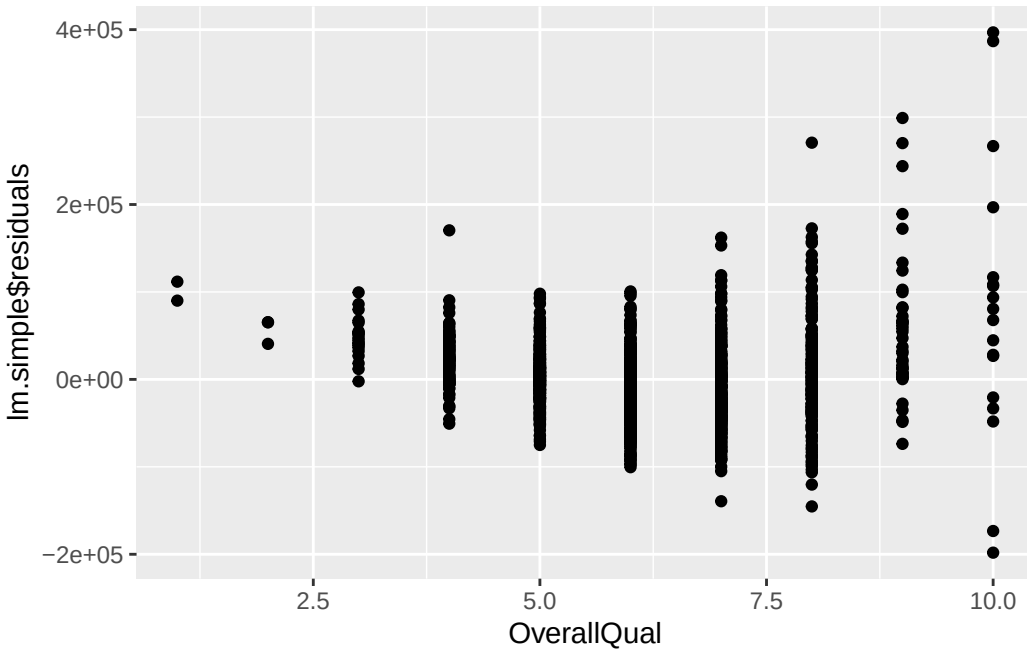

What do you notice about the QQ-plot and residual vs fitted plot? Looking at the Residuals vs Leverage plot, are there any noticeable outliers or high-leverage points? We can get the studentized residuals and leverage values using the following code:

```
stud.res = abs(rstudent(lm.simple))
lev = hatvalues(lm.simple)
```

Look at the rows in the dataset with high residual values or high leverage statistics (the class notes might be useful for how to define "high"). What do you observe?

Another useful plot is to look at residuals vs predictor plots. Here, we only have one predictor: OverallQual.

```
ggplot(data=house.sales) + geom_point(aes(x=OverallQual,y=lm.simple$residuals))
```




We observe that there is a trend of increasing residuals as OverallQual increases. This might indicate some non-linearity in the model. We test this hypothesis out by adding in a log(OverallQual) term to our model as follows:

```
lm.nonlin = lm(SalePrice~OverallQual+log(OverallQual), data = house.sales)
summary(lm.nonlin)
```

```
Call:
lm(formula = SalePrice ~ OverallQual + log(OverallQual), data = house.sales)

Residuals:
    Min      1Q  Median      3Q     Max 
-238558  -26157    -446   20080  356442 
```

```
Coefficients:
                 Estimate Std. Error t value Pr(>|t|)
(Intercept)        113129      18212   6.212 6.82e-10 ***
OverallQual         93577       4089  22.887  < 2e-16 ***
log(OverallQual)  -282438      23429 -12.055  < 2e-16 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 46380 on 1457 degrees of freedom
Multiple R-squared:  0.6596,    Adjusted R-squared:  0.6591
F-statistic:  1412 on 2 and 1457 DF,  p-value: < 2.2e-16
```

This corresponds to the model SalePrice $= \beta_0 + \beta_1 \text{OverallQual} + \beta_2 \log(\text{OverallQual}) + \epsilon$. Looking at the summary of the model, does the nonlinear term seem to significantly improve the model? Run through the diagnostics again and notice whether the trends we observed before are still there.

## Multiple Linear Regression

It looks like adding the non-linear term didn't really fix our problem and we might be missing some other information that will help better predict SalePrice. Let's see what happens when we throw all the variables into the model:

```r
lm.full = lm(SalePrice~.,data=house.sales)
summary(lm.full)
```

```
Call:
lm(formula = SalePrice ~ ., data = house.sales)

Residuals:
    Min      1Q  Median      3Q     Max
-401039  -15018    -752   13741  261507

Coefficients:
                      Estimate Std. Error t value Pr(>|t|)
(Intercept)         -1.165e+06  1.419e+05  -8.210 4.91e-16 ***
LotArea              7.585e-01  1.046e-01   7.252 6.72e-13 ***
NeighborhoodBlueste -2.159e+04  2.623e+04  -0.823  0.41061
NeighborhoodBrDale  -2.562e+04  1.245e+04  -2.057  0.03983 *
```

```
NeighborhoodBrkSide  1.310e+04  1.087e+04   1.205  0.22831
NeighborhoodClearCr  1.779e+04  1.144e+04   1.555  0.12013
NeighborhoodCollgCr  1.331e+04  9.093e+03   1.464  0.14351
NeighborhoodCrawfor  3.214e+04  1.073e+04   2.994  0.00280 **
NeighborhoodEdwards -1.209e+03  9.911e+03  -0.122  0.90293
NeighborhoodGilbert  1.985e+03  9.557e+03   0.208  0.83546
NeighborhoodIDOTRR   2.005e+03  1.150e+04   0.174  0.86162
NeighborhoodMeadowV -9.585e+03  1.244e+04  -0.771  0.44098
NeighborhoodMitchel  5.768e+03  1.018e+04   0.567  0.57100
NeighborhoodNAmes    9.106e+03  9.506e+03   0.958  0.33831
NeighborhoodNoRidge  6.577e+04  1.042e+04   6.313 3.66e-10 ***
NeighborhoodNPkVill -1.536e+03  1.465e+04  -0.105  0.91654
NeighborhoodNridgHt  7.228e+04  9.491e+03   7.616 4.75e-14 ***
NeighborhoodNWAmes   3.407e+03  9.806e+03   0.347  0.72829
NeighborhoodOldTown -2.302e+03  1.061e+04  -0.217  0.82820
NeighborhoodSawyer   9.526e+03  1.007e+04   0.946  0.34447
NeighborhoodSawyerW  6.061e+03  9.815e+03   0.618  0.53700
NeighborhoodSomerst  2.062e+04  9.359e+03   2.203  0.02775 *
NeighborhoodStoneBr  7.157e+04  1.108e+04   6.460 1.43e-10 ***
NeighborhoodSWISU   -4.460e+02  1.232e+04  -0.036  0.97114
NeighborhoodTimber   2.515e+04  1.049e+04   2.398  0.01663 *
NeighborhoodVeenker  4.042e+04  1.370e+04   2.950  0.00323 **
OverallQual          1.655e+04  1.180e+03  14.031  < 2e-16 ***
OverallCond          6.713e+03  9.371e+02   7.164 1.26e-12 ***
YearBuilt            5.640e+02  7.121e+01   7.919 4.77e-15 ***
GrLivArea            6.743e+01  3.429e+00  19.663  < 2e-16 ***
BedroomAbvGr        -7.062e+03  1.492e+03  -4.734 2.42e-06 ***
BathAbvGr           -4.738e+03  2.691e+03  -1.761  0.07851 .
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 34960 on 1428 degrees of freedom
Multiple R-squared:  0.8104,    Adjusted R-squared:  0.8063
F-statistic: 196.9 on 31 and 1428 DF,  p-value: < 2.2e-16
```

Here the "." indicates to add all the variables. This can be useful if we don't want to have to write out all the names. We can also use a "-" sign to ignore variables. For example, lm(SalePrice~.-BathAbvGr,data=house.sales) would use all possible predictors except BathAbvGr. Looking at the summary of this full model, try to answer the following questions:

How do you interpret coefficients for the different neighborhood variables? What is the baseline being used?

Are all the neighborhoods significant? How does this match what we observed earlier in the boxplots for SalePrice?

What do you notice about the bedroom and bathroom coefficients?

Is the BathAbvGr coefficient significant? Does this match your intuition?

How did the Adjusted $R^2$ value change from our previous simple model?

Although we achieved a much higher Adjusted $R^2$ value, we can also use an $F$-test to test whether there is a significant difference between the two nested models using the function **anova**.

```
anova(lm.simple, lm.full)
```

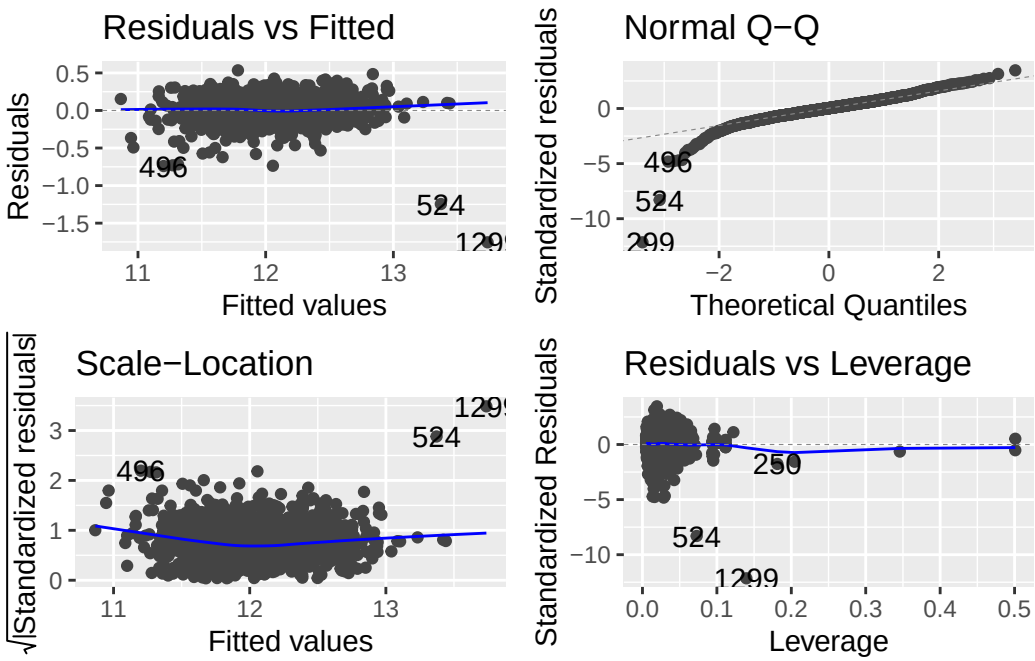
```
Analysis of Variance Table

Model 1: SalePrice ~ OverallQual
Model 2: SalePrice ~ LotArea + Neighborhood + OverallQual + OverallCond +
    YearBuilt + GrLivArea + BedroomAbvGr + BathAbvGr
  Res.Df        RSS Df  Sum of Sq      F    Pr(>F)
1   1458 3.4470e+12
2   1428 1.7456e+12 30 1.7014e+12 46.395 < 2.2e-16 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

This tests if the improvement in likelihood in the larger model is enough to be able to conclude that it is due to there being value in the additional covariates rather than overfitting to the noise in the data.

Run through the diagnostic plots again using **autoplot**. Even though we added in these other predictors, we still see that the residuals tend to increase as the predicted sale price increases. This indicates that we might want to use a log transformation for our response variable, which we thought might be useful earlier. Try the model with a log transformation for SalePrice. Does that seem to improve the fit?

```
lm.log = lm(log(SalePrice)~.,data=house.sales)
autoplot(lm.log)
```




We note that the trend in the residuals and the QQ plot do seem to have improved. Further, the Adjusted $R^2$ value is higher.

Last, we might consider adding interaction terms. For demonstrative purposes, we will add an interaction between OverallQual and GrLivArea. This corresponds to our intuition that an additional square foot of space should have a larger impact on price if the quality is higher. We add an interaction term by multiplying the two variables in the formula:

```
lm.int = lm(log(SalePrice)~.+OverallQual*GrLivArea,data=house.sales)
summary(lm.int)
```

```
Call:
lm(formula = log(SalePrice) ~ . + OverallQual * GrLivArea, data = house.sales)

Residuals:
     Min       1Q   Median       3Q      Max 
-1.42980 -0.07601  0.00616  0.08781  0.50540 

Coefficients:
                         Estimate Std. Error t value Pr(>|t|)    
(Intercept)             2.792e+00  6.291e-01   4.439 9.75e-06 ***
LotArea                 3.343e-06  4.599e-07   7.268 5.99e-13 ***
```

```
NeighborhoodBlueste   -1.605e-01  1.150e-01  -1.396 0.163017
NeighborhoodBrDale    -2.672e-01  5.472e-02  -4.883 1.16e-06 ***
NeighborhoodBrkSide    6.728e-03  4.797e-02   0.140 0.888487
NeighborhoodClearCr    1.035e-01  5.017e-02   2.062 0.039367 *
NeighborhoodCollgCr    5.168e-02  3.998e-02   1.293 0.196320
NeighborhoodCrawfor    1.571e-01  4.710e-02   3.336 0.000871 ***
NeighborhoodEdwards   -5.849e-02  4.384e-02  -1.334 0.182341
NeighborhoodGilbert    2.570e-03  4.195e-02   0.061 0.951160
NeighborhoodIDOTRR    -1.365e-01  5.078e-02  -2.689 0.007255 **
NeighborhoodMeadowV   -1.945e-01  5.500e-02  -3.537 0.000418 ***
NeighborhoodMitchel    1.477e-02  4.473e-02   0.330 0.741318
NeighborhoodNAmes      3.276e-02  4.186e-02   0.783 0.433909
NeighborhoodNoRidge    1.859e-01  4.632e-02   4.013 6.30e-05 ***
NeighborhoodNPkVill   -3.445e-02  6.434e-02  -0.535 0.592489
NeighborhoodNridgHt    2.255e-01  4.186e-02   5.386 8.40e-08 ***
NeighborhoodNWAmes     1.080e-02  4.304e-02   0.251 0.802004
NeighborhoodOldTown   -7.053e-02  4.668e-02  -1.511 0.131046
NeighborhoodSawyer     2.158e-02  4.437e-02   0.486 0.626717
NeighborhoodSawyerW    9.708e-03  4.310e-02   0.225 0.821813
NeighborhoodSomerst    7.225e-02  4.108e-02   1.759 0.078844 .
NeighborhoodStoneBr    2.256e-01  4.868e-02   4.634 3.91e-06 ***
NeighborhoodSWISU     -2.308e-02  5.407e-02  -0.427 0.669476
NeighborhoodTimber     1.007e-01  4.602e-02   2.189 0.028769 *
NeighborhoodVeenker    1.655e-01  6.007e-02   2.755 0.005947 **
OverallQual            1.401e-01  8.691e-03  16.115  < 2e-16 ***
OverallCond            5.545e-02  4.109e-03  13.493  < 2e-16 ***
YearBuilt              3.854e-03  3.131e-04  12.309  < 2e-16 ***
GrLivArea              5.377e-04  3.837e-05  14.016  < 2e-16 ***
BedroomAbvGr          -1.900e-02  6.737e-03  -2.820 0.004864 **
BathAbvGr             -1.842e-02  1.190e-02  -1.547 0.121980
OverallQual:GrLivArea -3.335e-05  4.769e-06  -6.992 4.14e-12 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 0.1533 on 1427 degrees of freedom
Multiple R-squared:  0.856, Adjusted R-squared:  0.8528
F-statistic: 265.1 on 32 and 1427 DF,  p-value: < 2.2e-16
```

Was the interaction term significant?

We could continue to build upon our model using the principles you have seen in class. For example, we could remove any insignificant variables, try coding neighborhood into a single 0/1 variable (great/non-great neighborhood), and add an interaction between being in a great

neighborhood and overall quality. If you have time, you can try out some of these changes on your own. Remember that when you write up your data analysis, you should justify your modeling choices along the way.